



Circularity in textile waste: Challenges and pathways to sustainability

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ABSTRACT

Emerging fast fashion concepts in the textile industry have contributed to the over-consumption of clothing and the consequent over-production of waste. Meanwhile, consciousness of the environmental pollution associated with the ongoing production and disposal of textiles has greatly expanded. For these reasons, greener and cleaner solutions are constantly being sought to remedy the negative impact of this waste on the environment sustainably. In this respect, the aim of this study is to provide an overview of the negative impact of textile waste on the environment, also the current research aimed at cleaner and more sustainable biotechnologies and to spotlight new developments in waste management strategies to remedy it. To achieve this, a literature review is provided based on data from Scopus, Web of Science and Google Scholar. The outcome of this study reveals that: i) the production of textile waste has increased significantly with the increase in clothing production; ii) the lack of rigorous environmental legislation related to the textiles production can further exacerbate the problem and contribute to increased environmental pollution; iii) the existing textile waste management programmes have proved unsuccessful; iv) clean and sustainable solutions such as biological treatment are required to overcome the harmful effects of textile waste; v) biotechnologies that use these wastes may generate valuable new products, clean energy, etc. Ultimately, overview of current-state-of-art suggests that textile waste can be successfully managed through biological treatment (using micro-organisms and especially enzymes), and thus mitigate the inappropriate textile waste management programs, thereby demonstrating the added value of the current study for the forthcoming research.

1. Introduction

For several million years, before humanity and technology developed, everything from the food we ate to the materials used would return to nature, but now nothing is completely recyclable; this could be due to the evolution of material technologies, which have become increasingly thorny. The concept of waste does not exist in nature, it is human production and consumption that establishes it. Thus, it becomes our responsibility to eliminate waste sustainably (Pathak, 2019). Waste management (WM) has become a global future challenge. Since antiquity, the textile industry has been considered one of the most valuable industries for humanity (Ütebay and Çay, 2020). The production of textiles and clothing becomes a daily necessity, thus contributing to the global economy. Nevertheless, this industry is one of the main industries that adversely affect human health and contribute to environmental pollution through resource scarcity and the clothing production process (Niinimäki et al., 2020). Furthermore, textile consumption is expected to continue to grow approximately 63% by 2030, implying the generation

of an increasing amount of textile waste, which is attracting worldwide attention (Ribul et al., 2021). According to Xu et al. (2022) this industry contributes to 20% of the total global waste.

In fact, the current pattern in production of the textile industry follows a linear economy: extraction/harvest, fabrication, delivery, utilization and then disposal in landfills or incinerators, depicting a tremendous waste of resources and energy (Schumacher and Forster, 2022).

Currently, the global economy is less than 10% circular, which is a waste of valuable resources (Fan et al., 2024). Reducing the world's textile waste requires new technologies to improve the recovery of valuable materials from their waste stream and improve their reuse for the benefit of consumers. Textile waste is today considered a precious resource, but unfortunately, the way we view it is obviously not enough, we just need to choose the right tools to make it a treasure. Recently, the requirement of a circular economy for the textile industry has been extensively advocated as a good solution to address the negative impacts of fashion production and consumption (Sandin and Peters, 2018).

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Circular economy is a developing concept; it implies that the resources used might soon be obsolete, so the idea is to know how to use natural resources and waste in a smart and sustainable way, to have a sense of reuse, sharing, repair, refurbishment, and recycling (Ribul et al., 2021).

In this regard, it is widely advocated that biological treatment could be used as a recovery method to transform textile waste into a valuable product instead of using chemical or even mechanical technologies that would otherwise contaminate the environment and degrade textile quality (Chopra et al., 2023). This waste should be considered useful, which perfectly coincides with the circular economy concept. Biological methods focus on resources that can be further converted into renewable nutrients, such as composting to produce biofertilizers that can regenerate soil health and improve plant growth or into clean energy (biogas) through anaerobic digestion (Kumar et al., 2020).

This investigation focuses on the concept of applicability of the circular economy to textile waste recycling. The study presents different challenges related to the problem of textile and clothing waste and how to deal with it in a clean and sustainable way, as well as the proposed, predominantly biological solutions to overcome this problem.

2. Methodology

This study gathers and synthesizes the data collected by academic researchers on the current situation of global textile production, as well as on textile waste, its impact on the environment, biotechnological strategies to remedy it and recommendations for the implementation of circularity in the textile sector. The literature review was carried out by searching the Web of Science (Clarivate Analytics) and Scopus database using the keywords ‘textile waste’, ‘global production of textile’, ‘impact of textile waste’, ‘textile waste circular economy’, whereby specific searches broadened the scope of this review article in quantitative terms. Over 9003 articles were retrieved in the extended search. The search was then refined by selecting only the most recent articles in the field. Ultimately, the search highlighted 120 scientific articles providing basic data on textile production, waste production, the impact of textile waste and biotechnological approaches to textile waste circularity. Approximately 70 articles were cited in this review; articles meeting the criteria below were included in the study:

The article review covers the following areas: i) Global challenges in textile production and waste generation, ii) Textile waste management strategies, iii) Environmental and human impacts of textile waste, iv) Guidelines for the promotion and achievement of circularity in the textile sector.

3. Growth of textile production

The term “textile” designates the underlying materials developed from cotton, jute, silk, wool, nylon, acrylic polyethylene, polyester, etc. (Schumacher and Forster, 2022), thus involving various processes; woven fibers and non-woven processes such as stitching are employed to use these materials to construct garments, clothing and other items (Table 1) (Kumar et al., 2020). These materials are plant- or animal-based, cellulosic, organic synthetic/semi-synthetic, and inorganic ones (Fig. 1) (Piribauer and Bartl, 2019).

Table 1
Global textile market.

Textile	
Type	Fabrics, yarn, fiber, thread, home furnishings and floor coverings, textile and fabric finishing, and fabric coating mills.
Process	Woven, non-woven.
Textile Application	Broad-woven fabrics, narrow fabric mills and schiffli machine embroidery, non-woven fabric, knitted fabrics
Outlook	home furnishings, floor coverings, textile and fabric finishing mills, fabric coating mills.

Over the last few decades, the demand for cotton, wool, and cellulosic products has remained roughly constant. At the same time, the demand for synthetic materials, particularly polyester, has increased dramatically due to outstanding performance properties and cost effectiveness (Kabir et al., 2019). In fact, 60% of clothing and 70% of household textiles are made from synthetic fibers, a figure that is expected to increase further in the future, then by cotton (24%) and finally wool (14%) (Khairul Akter et al., 2022). Today, 60% of the world’s fiber production is destined for the fashion industry (Schumacher and Forster, 2022), the rest is reserved for domestic, industrial, geo-, agro-, and hygienic textiles (Niinimäki et al., 2020).

Recently, textile production has expanded drastically, and this sector is a major contributor to economies around the world. According to Market Research’s report, the global textile business expanded from \$573.22 billion in 2022 to \$610.91 billion in 2023, at a compound annual growth rate (CAGR) of 6.6%. The textile market is predicted to rise further to \$755.38 billion by 2027, at a CAGR of 5.5%. Asia Pacific was the leading area in textile business in 2022, followed by Western Europe, which is the second most important. The global production of the sector worldwide is expected to exceed 99 million tons per year (Ütebay and Çay, 2020). These statistics reflect the significance of the industry’s contributions to the environment.

4. Textile waste production

The existing system for producing, supplying and wearing clothes operates in an almost entirely linear pattern. It is reckoned that well over half of all fashion and/or fast fashion manufactured is discarded in less than a year. Consequently, massive quantities of non-renewable resources are exploited to produce clothes that are frequently used only for a relatively limited period, after which the materials are mostly destroyed by landfill or incineration. Actually, the growing consumer interest in fashion and the trend towards the fast fashion have dramatically expanded the production and consumption of clothing (Johnson et al., 2020), leading to a substantial amount of waste at every stage of the manufacturing process, including spinning, knitting/weaving, dyeing, making-up and finishing. They encompass apparel and clothing, home and hospitality textiles, contract textiles, uniforms and workwear, upholstery fabrics, as well as excess inventory, dead stock, non-conforming product, damage and returns from manufacturers or retailers (Schumacher and Forster, 2022). This linear system creates untapped economic opportunities, puts pressure on resources, pollutes and degrades the natural environment and its ecosystems, and creates significant negative societal impacts at local, regional and global levels.

In this regard, it is expected that global textile waste will expand by 60% per year between 2015 and 2030, generating an extra 57 million tons of waste each year, to reach a total of 148 million tons per year. According to Khairul Akter et al. (2022), roughly 53 million tons of textile fibers are consumed each year, which leads to over 92 million tons of textile waste being produced each year worldwide, 73% of which is incinerated and/or discarded, which represents a waste of over USD100 billions worth of raw materials annually. The economic worth of these harmful externalities is tough to estimate, despite the latest fashion industry report indicated that the potential overall gain to the global economy might be around USD 192 billion in 2030 if the fashion industry tackled the environmental and societal impacts of the existing status (Shirvanimoghaddam et al., 2020).

In Bangladesh, (the world leader in textile production), it is estimated that roughly 577,000 tonnes of waste are generated by garment and fabric factories, of which nearly half of the total is 100% recyclable cotton waste that is worth around US\$100 million (Pavarini, 2021). In Cambodia, for example, the fashion sector is involved in roughly 60% of water pollution and 34% of chemical pollution (Angelov, 2016). Solid textile waste comprises biodegradable components such as fibres, yarns and fabric scraps (Khairul Akter et al., 2022). These materials can be recycled into new raw materials or even decompose to produce valuable

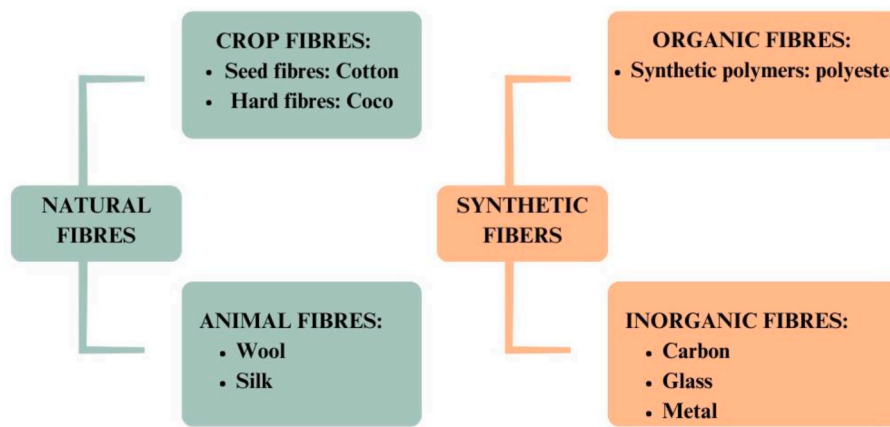


Fig. 1. Different fibre types.

products that could be used as biofertilizers and/or biogas (Yalcin-Enis *et al.*, 2019). However, garment industry's waste is a combination of natural fibers, synthetic fibers and other substances, such as metal zips, acrylic, wooden, or shell buttons and metal snaps used in apparel, rendering them challenging to degrade (Islam, 2021).

Furthermore, these wastes might be categorized as pre-consumer, post-industrial and post-consumer ones (Tang, 2023): i) Pre-consumer waste is incurring throughout the clothing and textile production process, it can also be described as the clean waste. It could be short fibers, combers, yarn waste, rejected fabrics due to manufacturing or dyeing faults, garments cutting waste, rejected garments, trimming, end lots of surplus production. In this sense, between 10% and 20% of the textiles are wasted and scrapped throughout the production process (Kabir *et al.*, 2019), ii) Post-industrial textile waste is designated as virgin or clean waste, which is released without being processed (Johnson *et al.*, 2020), iii) Post-consumer textile waste consists of all types of garments or textiles that are no longer used by the consumer, reflecting the larger stream that encompasses mixtures of all types and often harbors contaminants (Schumacher and Forster, 2022). Additionally, over the past few decades, the prevalence of pre- and post-consumer textile waste has expanded dramatically. The abrupt expansion of textile waste disposal has caused considerable concern for the environment and public health (Niinimäki *et al.*, 2020). Ultimately, the textile manufacturing industry has a significant footprint on the environment, whether it's the use of resources such as raw material, water and chemicals in the production of fibres, yarns and textiles, or the CO₂ emissions released in the fabrication, supply and consumption of garments.

5. Impact of textile waste

Changes in clothing consumption and consumer purchasing power, making textile sector one of the most prominent businesses in the world. But this essential activity involves critical human and environmental concerns at many stages of production, from raw materials to final production, including waste production (Ütebay and Çay, 2020). The textile and clothing industry's supply chain is lengthy and extremely complex, from agriculture and petrochemical industry (to make the fibers) to production, logistics and retailing (Niinimäki *et al.*, 2020). Although economically valuable, these industries are rated among the most polluting, producing enormous amounts of solid waste and wastewater. The textile and clothing industry has a harmful impact not only on public health, but also on the environment (waste generated, resources used, carbon footprint) along its entire supply chain (Sandin and Peters, 2018). It is a major contributor to man's adverse impact on the environment (soil, water, and air pollution), as it uses large amounts of water, energy, and chemical products for bleaching and dyeing processing (Schumacher and Forster, 2022) (Fig. 2), it is heavily dependent

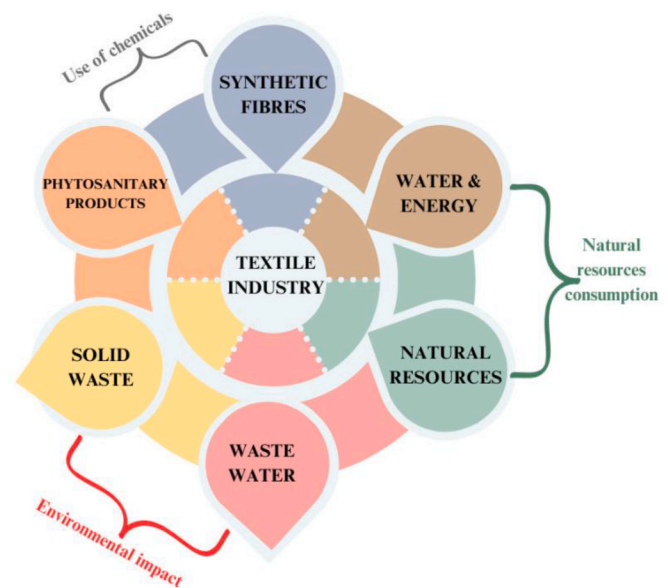


Fig. 2. Environmental impact of the textile industry.

on resources (natural fibers), as well as the use of pesticides to grow cotton (Saskia *et al.*, 2019). In fact, fashion, or even recently evolved fast fashion (known as single-use fashion) is the world's second biggest polluter, just behind oil and gas, causing considerable damage to the planet, exploiting workers and harming animals (Ellen MacArthur Foundation, 2017). It can be described as a toxic ecosystem of over-production and overconsumption.

In this regard, it is estimated that the consumption of clean water during textile production is exceeded only by agricultural activities, and water is used almost in all phases of textile production (washing, bleaching, dyeing, etc.) (Ütebay and Çay, 2020). For example, to produce 1 kg of textile, between 80 and 150 L of water are required, and each tone of cotton transformed into fabric consumes around 65 MWh of electricity and 25,000 L of water. Almost ¾ of the water is used for what is known as wet processes (Yaseen and Scholz, 2018). Additionally, approximately 4% of the world's freshwater consumption, or 93 billion m³ of water, is used annually by the textile sector (María Fleischmann, 2019). Cotton has the biggest water footprint of all fashion fibers and, as 44% of cotton is grown for export, roughly half of cotton's impact on local water use is attributable to foreign demand (Roos *et al.*, 2019). In fact, fashion facilities are a primary contributor to water pollution (Talouizte H *et al.*, 2013). About 20% of the world's wastewater is generated by this sector. For example, Bangladesh textile industry

produces around 56 billion liters of contaminated water a year, which can have a massive effect on the environment. Indeed, in Asia, over 70% of freshwater sources are polluted by waste generated by the textile industry (Yaseen and Scholz, 2018).

Another harmful impact of textile facilities is greenhouse gas (GHG) emissions. Such GHG emissions have been associated with 1.2 billion tons of CO₂-equivalent greenhouse gasses, which is more than that of all international transport (Niinimäki et al., 2020). Furthermore, CO₂ emissions are expected to expand by more than 60% to approximately 2.8 billion tons annually by 2030 (Sandin and Peters, 2018), making this industry a significant contributor to global warming. It is also considered that global output will be 2.9 billion tons of CO₂-equivalent emissions, of which two-thirds are related to synthetic compounds involved in fiber processing, textile manufacturing and apparel production. Conventionally cultivated cotton may emit 3.5 higher levels of CO₂ than organic cotton in the USA. Regarding the wool, the main contribution to gas warming pollution from its production is commonly enteric methane, which is generated through the digestive process of ruminants, is a potent greenhouse gas (Henry, 2012). Nevertheless, natural fibers have a significantly lower carbon footprint than synthetic ones (Fortuna and Diyamandoglu, 2017). In this respect, China's textile industry for example, relies on energy produced by coal-fired power stations, making its carbon footprint 40% higher than that of textiles produced either in Turkey or Europe (Pepermans, 2019).

Textile facilities have countless harmful effects, and thermal energy is one of the biggest concerns. Thermal energy consumption is required to heat water and dry materials in textile facilities, while in other manufacturing stages, such as spinning and weaving, it is predominantly electrical energy. Previously, it was reported that nearly 1 trillion kWh of electricity is required to produce 60 billion kg of fabric per year, which is a significant waste of energy (Table 2) (Ütebay and Çay, 2020). In this regard, the energy consumption varies according to fiber type - natural fibers (cotton, other than cotton celluloses, wool and hemp) require less energy but more water during production than synthetic fibers (polyester and polyamide). For example, polyamide processing requires 160 kWh per kg of fiber (Connell, 2015).

Whilst the textile industry continues to grow, the situation appears set to get considerably worse, but the use of water and energy is not the only reason why textile production can have an impact on the environment. In fact, textile manufacturing contributes to the emission of volatile organic compounds used in the production process. From inks and azo dyes to caustic soda, anthraquinones, phthalocyanines, salts, sulfuric acid, heavy metal lubricants, gas pedals, and solvents and all sorts of alkali and bleaching agents, most of the wet processes in textile production involve potentially harmful chemicals (Talouizte et al., 2013). For example, the fabrics are soaked into commercial bleaching agents and azo dyes, which ensure the color fastness of around 70% of textiles. Unfortunately, some of these dyes are contaminated with heavy metals such as cadmium, lead, chromium and mercury, which are the carcinogenic substances. In this sense, the textile industry relies on more

than 15,000 different chemicals throughout the manufacturing process (Niinimäki et al., 2020).

Indeed, untreated wastewater from the washing and drying processes flows unimpeded from the mills. This is also the case of textile made of wool materials. Wool requires treatment to eliminate soiling and parasites before it may be processed, which often entails chemical substances that can have harmful environmental consequences if improperly managed or simply discarded (Ellen MacArthur Foundation, 2017). Furthermore, 6% of the pesticides, 16% of the insecticides, 4% of the herbicides, and 1% of the fungicide production in the world is estimated to be delivered to cotton crops. However, most of these chemicals are harmful to employers because they could have a genotoxic and carcinogenic effects due to their persistence in the environment (Peaslee et al., 2020). In the environment, agrochemicals infiltrate the soil, where they impoverish biodiversity and soil fertility, interfere with biological processes and destroy microorganisms, plants and insects. The extensive exposure to agrochemicals may contribute to nausea, diarrhea, cancer, and respiratory diseases. Chronic pesticide poisoning is associated with nearly 1000 deaths a day, as well as neurological and reproductive health problems such as miscarriage, birth defects and infertility (Niinimäki et al., 2020). In Sweden, roughly 10% of the chemicals are considered to be potentially highly-risky to human health, especially azo dyes and acid dyes (Peaslee et al., 2020). Recently, Biyada and Merzouki (2024), described that these molecules are persistent and frequently detected at various concentrations in ground and drinking water, the soil, and the food chain. In this respect, around 60% of the materials used in clothing are plastics, including polyester, acrylic, nylon and other synthetic fibers, making them the largest known source of microplastic marine pollution, thought to be responsible for 35% of microplastic pollution in the world's oceans. Trillions of microfibers are emitted into the environment yearly through textile laundering, and the majority of them end up in the ocean (Baloyi et al., 2024). It is reckoned that, on current projections, there could be considerably more plastic than fish (by weight) in the oceans by 2050 (Shahid et al., 2024). The presence of plastics in the environment is a growing concern owing to their harmful effects on ecosystems and human health. Recently, plastic microfibers from the washing of plastic-based textiles such as polyester, acrylic and nylon have been recognized as a major contributor to environmental pollution (Ellen MacArthur Foundation, 2017). Furthermore, these synthetic materials are not biodegradable and can introduce toxic substances into surrounding groundwater and soil (Saskia et al., 2019).

In light of these fundings, there is a leakage of strict environmental and social legislation in the countries where production and manufacturing are carried out, further complicating the situation. In this regard, natural and/or synthetic materials are both involved in this negative outcome (Niinimäki et al., 2020). Upon degradation, natural cellulosic fibers produce methane, which is a strong greenhouse gas and contributes to global warming (Sandin and Peters, 2018). Ammonia produced from animal-derived textile waste, such as wool, may contribute to soil and water contamination (Schumacher and Forster, 2022). Because of their nonbiodegradable and toxic characteristics, petroleum-based synthetic textile wastes have a harsher effect on the environment. The leachate produced during the disposal of textile waste in landfills has a significant capacity to contaminate surface and groundwater sources (Ütebay and Çay, 2020). Therefore, it can be concluded that by producing fibers and thus discarding textile waste, we contaminate our environment in many ways while exhausting our scarce natural resources, which presents a new challenge to sustainability and threatens future generations. The immediate requirement to tackle the quantity of textile waste ending up in landfill sites has prompted researchers to investigate innovative and sustainable methods of recycling these materials and finding useful uses for them. The sustainability policy is intended to minimize resource consumption, waste production and pollution levels throughout the entire supply chain, from design to delivery (Tang, 2023).

Table 2

Rough fiber production, energy consumption, fresh water consumption and CO₂ emissions for cotton, polyester, non-cotton cellulose, polyamide, and wool according to (Niinimäki et al., 2020; Roos et al., 2019; Wang et al., 2017).

Materials	Fiber production (Million Tones)	Energy consumption (kWh/kg of fiber)	Freshwater consumption (L/kg)	CO ₂ emissions (kg/kg of fiber)
Cotton	27	48	1559	2.2
Polyester	54	108	21	3.3
Wool	1.2	120	530	17
Polyamide	6.1	160	40	8.3
Non-cotton cellulose	7.0	85	92	3.3

6. Textile waste management

Textile waste management has entered a critical phase, owing to the scarcity of adequate installations to treat and dispose of the huge amounts of these wastes produced in metropolitan cities (Yalcin-Enis et al., 2019). Instead, this waste could be used as a resource to generate added value products. This strategy stipulates that textile designers should consider improving sustainability, reusability and recyclability, and that textile production should strive to reduce related environmental and social impacts (Tang, 2023). Worldwide, the governments are attempting to reduce the quantities of waste dumped in landfill sites and increase the recycling of waste. Landfilling and incineration are two well-known methods of treating hazardous solid textile waste. The results showed that incineration has a lower impact on human health (28% of diseases compared with 46.8% for landfill), but a greater impact on ecosystem quality, meaning that both methods are unsustainable (Tisserant et al., 2017). To accommodate this, cleaner, more sustainable waste management practices should be adopted with more appropriate treatment technologies.

In this sense, most of the literature on the material waste domain in textiles and apparel are research outcomes on ways to recover, reuse and recycle waste materials efficiently and cost-effectively, which lines up perfectly with the concept of circular economy (Juanga-Labayen et al., 2022). The circular economy approach is designed to lengthen the life of materials and boost recycling and recovery with the intention of recognizing and maximizing the material service per resource employed, while minimizing environmental impacts and resource uptake (Tisserant et al., 2017), it likewise incentivises the gathering of products and their recovery within the same product chain (Yaseen and Scholz, 2018). In this regard, the 5 R approach (reduce, refuse, reuse, recycle and recover) to waste management has emerged internationally as one of the cornerstone concepts of the circular economy for a sustainable society (Tisserant et al., 2017). Researchers suggest using eco-friendly raw materials dyeing (Islam, 2021) and finishing processes (e.g., reduced water dyeing and waterless dyeing) to reduce materials waste in fabric processing (Fortuna and Diyamandoglu, 2017). Furthermore, most textile facilities are adopting recycling as one of the options for properly handling the textile waste generated. Recycling waste management is hampered by mechanical treatments of materials in generally. As the following sections illustrate, these are critical areas where knowledge is lacking, and further research is required to turn into clean, sustainable technologies.

6.1. Textile wastes and recycling

The environmental challenges arising from the growing accumulation of non-biodegradable textile waste, combined with the reduction of landfill areas, have led the textile and fashion sectors to consider and implement the environmentally-friendly strategies at a certain stage in the life cycle of textile products (Khairul Akter et al., 2022).

One of the main foundations of the circular economy is recycling. In general, recycling is the conversion of worn clothing, textiles, and waste into new commodities. Ideally, if almost all fashion textiles are recyclable, there should be no waste in the textile and clothing sector. Although the cost of recycling textile waste is low, the recycling rate is still relatively low, which can be explained by the variety of fibrous waste and its structure. Ütebay and Çay (2020), estimated that up to 95% of textile waste could be recycled into a variety of worthwhile products. Moreover, it is reckoned that over 60% of all clothing retrieved could be reused, 35% could be turned into cleaning rags and fiber recycling purposes, thereby dropping just 5% of clothing for disposal. Nevertheless, a significant portion of textile waste still ends up in landfill.

There are numerous approaches to tackling the waste, including minimizing overproduction and overconsumption, lengthening the life of products, and engineering products for improved circularity. The

process of recycling natural textiles involves sorting materials by type and color, minimizing the requirement of redipping the fabric, saving energy, water, and preventing pollution (Cesaro et al., 2019). The ever-increasing energy consumption and environmental issues are among the major challenges facing the world. One way to overcome these threats is to increase the use of recycled materials and environmentally friendly manufacturing methods. Recycling of textile waste can be mechanical, chemical, thermal, or a combination of these processes (Subramanian et al., 2020). It can also be classified according to the process and the final products.

Furthermore, recycling may depend on the disassembly process (Fig. 3), this process was thoroughly explained by Sandin and Peters (2018); Or even depending on the output, it could be recycled during the production process, which is linked to the development of new textile-like commodities. Alternatively, fiber-to-fiber recycling, which is carried out within the framework of the transformation of textile waste into new fibers that are further exploited to develop new clothing or other textile products, may be used.

In this sense, in mechanical textile recycling operations, all materials, hazardous or not, remain in the material and are introduced into the new product. Similarly, the mechanical recycling of textile materials leads to fibers of poorer quality, length, and toughness; ultimately, mechanical recycling can be considered as downcycling (Zandberga et al., 2023). In addition, mechanical recycling may be possible in the case of the transformation of waste into a new product of equal or higher quality than the original, which can be defined as upcycling (Yalcin-Enis et al., 2019). For example, in the case of mechanical recycling of cotton from fiber, it is feasible only for 100% cotton textiles. Therefore, recycled fibers should be mixed with virgin fibers in the production of new textiles, which is again a waste of resources (Sandin and Peters, 2018). Furthermore, according to Moorhouse and Moorhouse (2017), approximately 30% of recycled cotton fiber goes into new cotton garments, while up to 50% of recycled denim goes into denim products, through 50% recycled cotton also occurring in blended threads, in which cotton is mixed with synthetic fibers (PET) and/or recycled nylon. Furthermore, in the case of blended fibers (polycotton), another operation is mandatory to separate cotton from PET (Palme et al., 2017), which can be accomplished chemically by depolymerizing or dissolving one component while retaining the other, leading to a harmful impact on the environment and human health (Sandin and Peters, 2018).

Today, acoustic pollution is one of the largest global threats in the

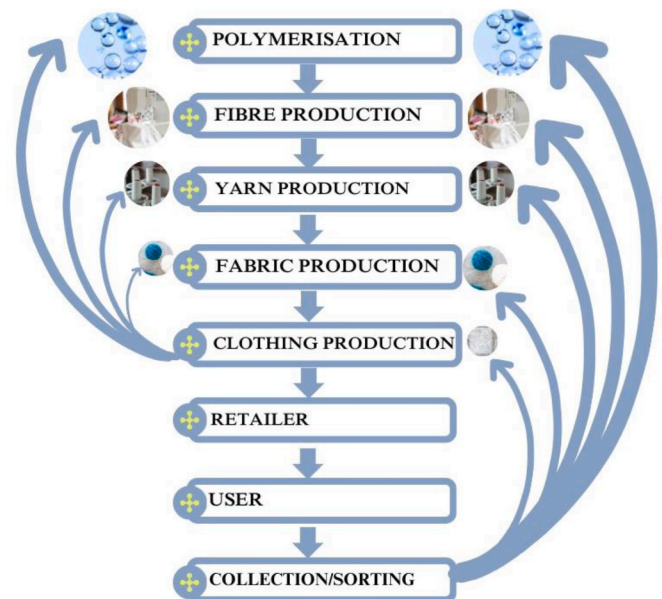


Fig. 3. Textile reuse and recycling routes.

world, as it affects both human health and the quality of the environment (Dissanayake et al., 2021). To address this problem, sound barriers appear to be a successful solution. Normally, acoustic materials, such as mineral wool and petrochemical polyurethane foams that are currently used, are not sustainable with respect to energy consumption, greenhouse gas emission, are not degradable and have human health impacts (Zach et al., 2016).

Besides, another alternative is recycling beyond the production process, which involves the manufacture of different commodities, such as biocomposite material for insulation (Massoudinejad et al., 2019). Many researchers around the world are involved in the development of insulation manufactured from natural fibers and/or waste materials (Zach et al., 2016). In addition, recycled insulation materials are established as a sustainable and affordable alternative to conventional synthetic or natural materials. Since recycled materials require significantly less energy and non-renewable resources, they conserve resources efficiently (Massoudinejad et al., 2019). In this regard, the recycling of textile waste as insulation material for automotive upholstery and construction could be a great illustration of the design of a new circular economy model and contributes to the protection of the environment and the minimization of energy consumption in buildings; it is an extremely cost-effective and sustainable recycling process (Ribul et al., 2021). In fact, due to the physical properties of textile waste, this material could be compatible with conventional building insulation materials (Islam, 2021). In this regard, Dissanayake et al. (2021), have conducted the implementation of cotton and polyester blends as insulation materials due to their superior properties and sound insulation characteristics. Implementing high quality insulation materials (thermal and acoustic) from natural or secondary fibers obtained from textile waste rather than for insulation of the construction envelope and/or interior structures is crucial, in terms of minimizing the adverse impact on the environment, reducing energy consumption (the energy necessary to heat and cool our buildings and vehicles can be reduced), landfill area, natural resources exhaustion, greenhouse gases, pollution (noise, air, water, soil); and preserve the health of humans and biota (Baloyi et al., 2024). Recycling textile waste into insulation materials also has potential upside for economic sectors, as it allows the sale and purchase of valuable scrapped fibers and thus produces low-cost insulation materials. Furthermore, the use of readily renewable natural materials considerably minimizes the consumption of non-renewable resources that are today involved in the manufacturing of insulating materials (Zach et al., 2016). Ultimately, despite the fact that recycling is well developed in terms of process, it cannot continue infinitely since it involves a deterioration in the quality of the fabric. Consequently, the majority of recycling technologies are downcycling procedures, as the final value of the products is lower. For this reason, cleaner, innovative and sustainable strategies need to be sought to remedy the unsatisfactory results achieved by recycling. In this review, we attempt to explore the most appropriate and sustainable methods for dealing with the challenge of textile waste. We intend to spotlight the most innovative and effective way of processing textile waste to develop cleaner and more sustainable products (Yalcin-Enis et al., 2019). For this reason, the biological treatment of textile waste has been targeted.

6.2. Biological treatment of textile waste

Biomass is a biological material that is a renewable, potentially sustainable, and relatively environmentally friendly source of clean energy and materials. The use of biomass waste as resource materials not only provides a sustainable approach to cope with the growing demand for resource materials, but also to solve waste recycling issues.

The intensive use of natural and synthetic resources by the textile industries, combined with the adverse impact of these activities on the environment and human health, has drawn the attention of researchers. Several environmental studies have been conducted to address these concerns by evolving to a circular economy for textiles, integrating

treatment and recycling technologies to protect wealth, the environment and public health (Zandberga et al., 2023). Due to its organic composition, textile waste is considered a treasure trove of nutrients for the growth of microorganisms, which makes biological treatment with microorganisms and their enzymatic systems one of the most recommended treatments for this type of waste. In this regard, biological technologies could be used as a clean approach to chemical processes, due to their advantages, such as limited energy consumption, environmentally friendly processes, and the use of renewable materials (Khan et al., 2023) (Table 3).

In this section, several ways of treatment will be discussed: anaerobic digestion, composting, bioaugmentation, fermentation, enzymatic depolymerization, etc.

6.2.1. Anaerobic digestion

Turning waste into clean energy is one of the most innovative approaches to the biological treatment of textile waste. In fact, more than just an area of innovation, it's an integral part of a successful circular economy, in which waste is repurposed to continue supporting human life in other ways. Biofuels generated using industrial lignocellulosic feedstocks such as textile waste, agricultural, and municipal wastewater, commonly containing a high amount of organic matter, are a promising source of clean energy, such as biogas (biohydrogen, biomethane) and bioethanol, etc., instead of fossil fuels (Khalid et al., 2011). The enormous amount of solid waste generated by the textile industry represents a veritable environmental threat. This waste has an extremely high moisture content and an organic load. Due to its richness in fats, proteins and carbohydrates, textile waste is considered an important feedstock that could be used to generate high quality renewable and clean fuels, such as biogas. Anaerobic digestion is stimulated by bacteria and then the methanogenic archaea in the absence of oxygen to produce a gaseous mixture of methane (biogas) and CO₂ from organic matter (lipids, proteins, and carbohydrates) (Opwis and Gutmann, 2012). Using bioenergy production, the most important benefit is to maximize utilization of underutilized resources, reduce waste volume, disposal of negative effects, create new opportunities for integrating nutrient recycling and clean energy production, and can also help to reduce CO₂ emissions and substitute chemical fertilizer using the digest produced during biogas, which could be used as a source of organic fertilizer (Jeihanipour and Taherzadeh, 2009). The generation of bioenergy is a viable approach to reduce the environmental effect of waste from the textile industry. Biogas can be used to generate green energy; heat, electricity, and renewable transportation fuel (Kumar et al., 2020).

6.2.2. Composting treatment

Composting is rated as one of the most sustainable and relatively inexpensive organic waste management strategies and is an aerobic process that consists of mineralization and humification of organic matter by microorganisms to a valuable final product; compost. Compost is a nutrient rich biofertilizer that can further rehabilitate soils through its soil enrichment qualities. It could be considered as low-priced organic ingredients compared to chemical fertilizers due to their high concentration of nutrients, especially K, P, N, and their potential to mobilize soil ions and simultaneously dispose of these wastes in an environmentally friendly way (Cesaro et al., 2019). They include a higher plant-available water holding capacity (WHC) and cation exchange capacity, a lower bulk density, pH stabilization, and better aggregation.

It is widely recognized that textiles and clothing made from natural materials such as cotton, jute, silk, and wool are rich in cellulose and lignin as main components. Previously, the abundance of aromatic compounds, phenolic, peptide, aliphatic, and polysaccharide structures in the textile waste has been demonstrated. Moreover, because of their outstanding contribution to composting, microorganisms have arisen the scientists' reflection. Biyada et al. (2022), have conducted a study to identify the microorganisms (bacterial and fungal) involved in textile

Table 3
Open-loop biological pathways in which textiles are the end product of fermentation or in which fermentation produces amino acids and enzymes.

Method	Feedstock used	Product	Reference
Composting	Wool, Cotton, Synthetic cellulosic materials (jeans).	Bio-fertilizer	Biyada and Merzouki (2024)
Anaerobic digestion	Wool, Cotton, Blend Cotton/PET, Cellulose.	Biogas (methane, CO ₂)	(Jeihanipour and Taherzadeh, 2009; Kumar et al., 2020; Ribul et al., 2021)
Enzymatic depolymerization	Blend Cotton/PET, Jeans, Wool, Cotton, PET,	Glucose, amino acids, PET fibers, terephthalic acid (TPA) and ethylene glycol (EG),	(Al-Sabagh et al., 2016; Chopra et al., 2023; Ribul et al., 2021)
Fermentation	Wool, Cotton, Cotton/PET Blend, Cellulosic, Jeans.	bioethanol, citric acid, succinic acid and butanol, bacterial cellulose, silk proteins	(Jeihanipour and Taherzadeh, 2009; Kumar et al., 2020; Ribul et al., 2021)
Bioaugmentation	Wool, Cotton, Cellulosic, Jeans.	Bio-fertilizer & Bioenergy	Rodrigues et al. (2024)

waste by composting using next generation sequencing, as well as their enzymatic activities (cellulase, glucan 1,4- α -glucosidase, endoglucanase, endopolygalacturonase, β -xylanase, etc.) using the UniProt database. This study provided a database of the most outstanding microorganisms involved in textile waste, which can be further exploited as accelerating agents in the degradation process, in order to shorten the composting time. Due to the prevalence of recalcitrant molecules, composting time could be long, which is a disadvantage for this kind of treatment. Therefore, further studies on textile waste degradation through the bioaugmentation using selected autochthonous strains based on this database may solve the concerns about the composting period. Bioaugmentation could be considered as an innovative alternative to the conventional composting process, which will be used in the future to further improve the composting process involving different types of industrial waste. Ultimately, this study proved that composting can be an exceptional approach for biotechnologists to explore the potential of microorganisms and their enzymes that capable of breaking down cellulosic complexes of the waste. This area of research is relatively virgin and/or unexplored, which requires more in-depth research in the future.

6.2.3. Enzymatic depolymerization of textile waste

Biotechnologies such as fermentation and enzymatic processes are being further explored for the biological recycling of textile waste due to their high selectivity and inoffensive nature, but also owing to their low energy requirements and their renewable sources ([Jeihanipour and Taherzadeh, 2009](#)). In this regard, biomass could not only be reused as a biofertilizer, but also be converted into clean bioproducts such as sugar, cellulose, and alcohols. Recently, the use of enzyme complexes to degrade textile waste, especially those of natural origin, into high-value byproducts that can be reused has been widely recommended ([Baloyi et al., 2024](#)). The upside of using enzymes to break down textiles is that they can overcome problems related to the heterogeneous nature of textiles, such as polycotton ([Ribul et al., 2021](#)). [Table 4](#) illustrates recent advances in the field of micro-organisms and their enzymatic activities, which are well known for their ability to cleanly recover textile waste. Previously, [Jeihanipour and Taherzadeh \(2009\)](#), reported the use of selective enzyme complexes to depolymerize a cotton-polyester textile waste mixture into glucose and thus its fermentation by *Saccharomyces cerevisiae* into ethanol, which is today the most significant renewable fuel produced from solid waste of cellulosic nature. In the case of wool biomass, the processing is a bit different since it is made up of a fibrous polymer of amino acids called keratin ([Fang et al., 2013](#)). Additionally, [Fakhfakh et al. \(2013\)](#), reported the use of protease enzymes called keratinases produced by *Bacillus pumilus* to ferment wool polymers into monomers of amino acids. Furthermore, [Petek and Marinšek Logar \(2021\)](#), have found that the most extensively explored keratinolytic microorganisms belong to the genera *Bacillus*, *Chryseobacterium*, *Serratia*, *Stenotrophomonas*, *Aspergillus* and *Penicillium*.

Polyethylene terephthalate (PET) is the most commonly used polyester plastic, with nearly 70 million tons produced worldwide each year for textiles and packaging, this polymer is deployed to fabricate products such as polyester fibers ([Wei et al., 2019](#)). Plastic waste (synthetic

Table 4
Selection of recent novel and well-known micro-organisms and enzymes able to recover textile materials.

Microorganisms	Enzymes	Materials	References
<i>Bacillus (megaterium, thuringiensis, etc.)</i> <i>Penicillium (chrysogenum, corylophilum, etc.)</i> <i>Aspergillus (clavatus, oryzae, etc.)</i> <i>Trichoderma reesei</i> <i>Aspergillus niger</i>	Keratinases Cellulase Cellulase	Wool Cotton/PET Pure cotton, Jean and Cotton/PET	Petek and Marinšek Logar (2021) Wang et al. (2018) Hu et al. (2018)
<i>Saccharomyces</i>	Cellulase	Pure cotton, Jean	Jeihanipour and Taherzadeh (2009)
<i>Penicillium sp.</i>	Cellulase, Cutinase, keratinas	Silk, wool	Mazotto et al. (2021)
<i>Pseudomonas</i>	Cutinase, Cellulase	Cotton and linen	

materials) is an ecological challenge, and enzymatic degradation provides a potentially green and upgradeable route for recycling polyester waste methods ([Subramanian et al., 2018](#)). This concern has attracted the attention of biotechnologists. In this regard, the biological approach to catalyze the hydrolysis of polyethylene terephthalate (PET) appears to be a cost-effective and environmentally friendly alternative to the chemical approach to PET depolymerization, which could contribute to alleviating the threat impact of PET production on the environment and allowing the sustainable development of a new breed of environmentally friendly materials ([Salvador et al., 2019](#)).

In this sense, the enzymatic depolymerization of PET in its monomer form (terephthalic acid (TPA) and ethylene glycol (EG)) has recently been demonstrated using distinguishable microbial polyester hydrolases such as *esterases*, *lipases*, and *cutinases*, using microorganisms cultured on PET as a source of carbon ([Tournier et al., 2020](#)), which may therefore be used for the reconstruction of new PET materials ([Al-Sabagh et al., 2016](#)). Microorganisms may be employed to further develop the constituent monomers from renewable sources. The partnership of biodegradation and biosynthesis would allow for a completely circular bio-PET economy, exceeding traditional recycling processes ([Salvador et al., 2019](#)). In this respect, several microbial species are identified, such as *Thermobifida fusca*, *Thermomyces insolens*, *Escherichia coli*, *Fusarium solanipisi*, and *Ideonella sakaiensis* strain 201-F6, which are known to grow in PET as a sole carbon source and therefore produce a broad range of thermostable polyester hydrolases, such as *esterases*, *cutinases*, *PET hydrolase*, etc ([Wei et al., 2019](#)). [Ribul et al. \(2021\)](#), reported the use of a newly developed enzyme, namely polyethylene terephthalate hydrolase or *PETase*. Recently, [Wei et al. \(2019\)](#), discovered another thermostable enzyme known as FAST *PETase*. The thermo*PETase* enzyme showed a wide working temperature range (30–60 °C) and considerably higher enzymatic activity than *PETase*. These enzymatic

activities could degrade this polymer into monomers, which can be used as raw material for new polymers (Chopra et al., 2023). Furthermore, depolymerization approaches such as glycolysis, methanolysis, hydrolysis, aminolysis, and ammonolysis have recently emerged as crucial strategies for the recycling of PET waste (Salvador et al., 2019). Glycolysis is a pathway from PET to glycols at high temperatures using catalysts such as metal acetates. Glycolysis has a significant advantage over other methods, as it allows recycling of colored and opaque PET, which cannot be recycled in any other way due to the presence of pigments (Tournier et al., 2020). Ultimately, plastic waste notorious for its non-biodegradability could be definitively recycled using the enzymatic activities of microorganisms, but this area of research should be further expanded by biotechnologists, to gather more in-depth insight into how this waste behaves during processing.

6.2.4. Fermentation treatment of textile waste

Fermentation is a metabolic process that involves the action of microbial species, such as bacteria and fungi, as well as other eukaryotic cells, through their enzymes on an organic substrate for develop useful products to humanity. Since antiquity, fermentation has been used to produce a wide range of enzymes, in particular, cellulases, hemicellulases, lignin peroxidases, proteases, lipase, etc., using microorganisms for different biochemical processes (Rathinamoorthy and Kiruba, 2022). These enzymes are involved in different processes such as depolymerization of polymers into monomers, as well as in the treatment and valorization of solid waste, especially those rich in cellulose, which is the case of textile waste. In this regard, several authors have reported the use of several species (fungi and bacteria) in textile depolymerization (Chopra et al., 2023). Ellilä et al. (2017), used textile waste as a growth medium for *Trichoderma reesei* to produce cellulases. In general, textile waste acts as a growth substrate for microorganisms to produce cellulases that are further exploited to depolymerize textile waste into the soluble monomer glucose fraction and, therefore, fermented to produce monomers such as lactic acid, which can subsequently be polymerized into polylactic acid fibers (PLA) that is employed to manufacture a new textile material, or even into other valuable inputs, especially clean products such as bioethanol, citric acid, succinic acid and butanol (Jeihanipour and Taherzadeh, 2009).

Recently, Rathinamoorthy and Kiruba (2022), have reported the production of cellulosic textile polymers (cotton, silk, amino acids and polyester) using a biotechnological approach of microorganisms, which could be a crucial step in recycling textiles by producing a new valuable textile rather than using natural resources. In this regard, several bacterial species are identified as cellulose producers, among which; *Acetobacter hansenii*, *A. pasteurianus*, as well as *Komagataeibacter xylinus* (Ribul et al., 2021). Bacterial cellulose has been widely used in packaging as well as in the medical industry as a dressing; its applicability has also been explored as an innovative alternative to vegetable leather. Bowen et al. (2018), have reported the use of *S. cerevisiae* and *E. coli* as silk proteins, which can be further exploited to produce alternative and sustainable protein-based textiles. Furthermore, several bacterial species are notable for their ability to assimilate and accumulate polyhydroxyalkanoates (PHAs) as cytoplasmic granules upon nutrient exhaustion, which can be subsequently processed into threads. Therefore, due to its biodegradability, PHA could be deployed to produce bioplastic and biobased textiles (Huang et al., 2017).

At the end, it is clear that the challenge of recovering solid textile waste into added-value products, and the necessity of adopting a coherent life-cycle approach to avoid unintended consequences, requires selecting the appropriate recovery pathway, from raw materials to end-of-life production, in order to avoid the accumulation and harmful impact of textile waste on the environment.

6.3. A new textile economy system is an attractive vision of a system that works

The overall approach to a new textile economy is that it is consistent with the philosophy of a circular economy: an economy that is inherently rehabilitative and regenerative, delivering benefits to business, society and the environment (reducing GHG emissions, minimized consumption of virgin materials and non-renewable energy, reducing plastic in the oceans, minimizing the pressure on water in water-scarce area, etc.) (Moorhouse and Moorhouse, 2017). In such a system, garments, textiles and fibers are preserved at their highest value during use, and reintegrated into the economy after use without ever becoming waste (Ellen MacArthur Foundation, 2017). Implementing this vision of a new global textile system hinges on several fundamental objectives: to gradually withdraw hazardous materials including microfiber discharges; to transform the way clothes are designed, sold and used in order to move away from their increasingly disposable nature; to radically improve recycling by transforming the design, collection and reprocessing of garments; and to use resources efficiently and switch to renewable energy sources (Fig. 4) (Khairul Akter et al., 2022).

In this regard, waste-free fashion design has been a niche market for many years, but new requirements for more sustainable trends in the fashion industry and low-waste production are propelling it to the forefront of creative innovation (Chopra et al., 2023). Designing, producing and providing access to higher-quality garments through new innovative business models are crucial if we are to move beyond the current linear pattern of clothing and would contribute to changing the perception of clothing from a disposable item to a sustainable product and therefore have improved products that last longer and create less waste (Khairul Akter et al., 2022). The process of switching to alternative closing production innovative business models will obviously depend on regions, income levels and current trends in apparel consumption.

In this respect, many brands are embedding zero waste into their business model and reusing their own textile waste as a raw material. This concept embodies a first-order innovation that goes beyond the closed loop to the infinite loop - or loop lessness. Not only is this a more economical manufacturing approach, it also provides a tighter link with textile mills as part of the creation of a circular economy, enabling brands to become more involved in fabric design, leading to greater exclusivity and innovation (Rissanen, 2011).

In the case of microfibers, although its collection has been the primary focus of attention to date, the removal of them by redesigning and engineering garments has attracted considerably less attention. As industry becomes increasingly aware of the problem of microfiber pollution, there are many research and deployment opportunities to build on to create new materials that take microfiber emissions into account from the outset or alternatively, develop production processes that prevent the further downstream loss of microfibers (International union for conservation of nature, 2017). This will presumably entail conceiving completely new materials from scratch, which are biodegradable or do not lose their microfibers, and which have the properties required for high-performance processing. In a circular textile system, the discharge of plastic microfibers should be excluded and/or its harmful effects eradicated (Ellen MacArthur Foundation, 2017). To overcome this, several brands are committed to the zero-waste concept. For example, brands like Adidas have launched the world's first sports shoes made entirely from ocean plastic. This concept enables Adidas to eliminate virgin plastic from its supply chain and develop shoes made from Bio-steel fibre, which replicates natural silk and is entirely biodegradable. A further initiative is the Davy J project, a new swimwear brand that seeks to establish that it is feasible to design competitive, high-quality products that are produced in a sustainable manner (Rissanen, 2011). The project promotes a circular economy by using 100% regenerated nylon yarn from consumer waste, including used fishing nets (Moorhouse and Moorhouse, 2017).

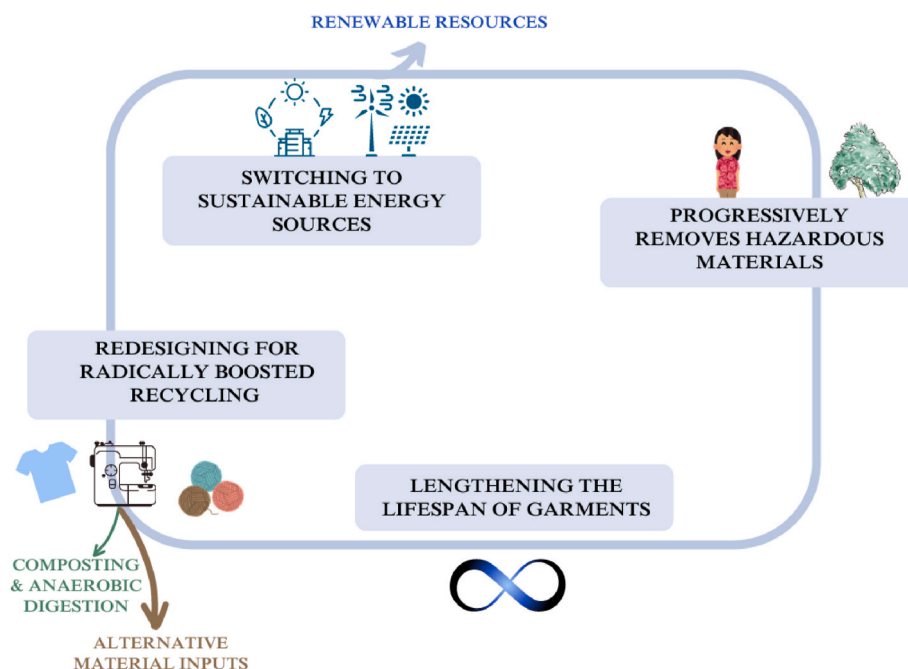


Fig. 4. A new textile economy approach.

Furthermore, boosting the number of times clothes are worn could be the most cost-effective way of capitalizing on value, reducing pressure on resources and lowering negative impacts. For example, if the number of times a garment is worn were doubled on average, GHG emissions would be reduced by 44%. Worldwide, customers waste up to US\$460 billions annually by discarding clothes they could otherwise continue to wear (Ellen MacArthur Foundation, 2017). These innovative new business models have led to three key actions that could help us move away from the disposable clothing culture: expanding short-term garment rental, boosting the appeal of sustainability, and expanding garment use through brand commitment and policy.

Ultimately, waste management and prevention could only be guaranteed if there is an awareness of the deleterious effects of the waste produced on the environment and on human beings, and of the significance of waste reduction and product reuse (Nielsen and Schmidt, 2014). Therefore, if we intend to overcome the major challenges confronting the planet today, we have to embrace innovation and experimentation (Khairul Akter et al., 2022). We need to move forward beyond natural resources and seek alternative ways of producing clean energy. A new textile economy inspired by the principles of the circular economy would enable us to achieve improved outcomes. To this purpose, governments and public authorities should undertake precautionary measures by applying the circular economy philosophy before a material, substance, or product reaches the point of becoming waste, thereby reducing the quantity of waste, by recycling products and lengthening of their lifespan, reducing the adverse effects of the waste produced on the environment and human health as well as the content of noxious substances in products (Yalcin-Enis et al., 2019).

7. Conclusions

This study describes the concept of the use of the circular economy to deal with the negative impact of textile waste on the environment (pollution, waste and natural resources deployed) and public health. Existing and forthcoming textile waste management technologies were reviewed during this investigation. This investigation criticizes the fact that the majority of studies focus on the recycling (mechanical and chemical) of textile waste as a solution to overcoming the harmful impact of these wastes, whereas biological treatments offer us reliable

and clean solutions for harnessing the richness of this waste as organic matter that could be reused to make new products instead of simply being recycled into a lower-quality product. The novelty of this study is therefore to highlight the predominant contribution of biological technology to the development of new, clean and valuable products from textile waste, in order to mitigate its potentially adverse impact. In this regard, because of their tremendous capacity to biodegrade a broad range of toxic compounds and thus biosynthesize nontoxic compounds that could be subsequently used as biofertilizers, clean energy, biocomposites, or even new virgin textile materials, microorganisms and their enzymatic mechanisms were the main target of this review. The results of this study revealed that the current conventional management of textile waste offers unreliable results, whereas biological treatment appears to be the solution, as it achieves very satisfactory results. To date, several micro-organisms have been identified for the recovery of textile waste, but knowledge of these micro-organisms, and in particular their enzymes mechanism of action, is still insufficient. We consider that the world microorganisms and their enzymes is infinite, therefore further research is required to produce a new enzyme capable of recovering textile waste and transforming it into clean and sustainable products. Ultimately, this review is a reliable and precious guide for the future researcher seeking to explore the promotion of cleaner, more innovative and sustainable technologies used to overcome the overproduction of textiles and their negative impact on the environment, thus contributing to the achievement of sustainable development goals.

CRediT authorship contribution statement

Saloua Biyada: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jaunius Urbonavičius:** Writing – review & editing, Validation, Supervision, Resources, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationship that could be construed as a potential conflict of interest.

Data availability

Data will be made available on request.

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